



Chapter 8 Factor Analysis

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§ 8.1 Introduction

- Conditions of principal components :
 - (1) higher accumulative contribution rate of first several principal components ;
(usually easy to satisfy)
 - (2) Reasonable and meaningful explanation of principal components
(sometimes difficult to satisfy)
- The application of factor analysis is similar to that of principal component analysis. It is also a dimensional reduction method. Since factors are easier to explain, it has higher success probability and wider application area.

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- In terms of methods, factor analysis is more exquisite than principal component analysis, theoretically more sophisticated. Principal component analysis only involves common linear transformation and don't involve model. The assumption of the existence of secondary moment is enough for principal component analysis, while factor analysis need set up a model with further assumptions.
 - Fundamental of factor analysis is to divide variables into groups with correlation in order to make same group's variables have higher correlation and different groups' variables have lower correlation. Each group variable represents a basic structure explained by common factors.
 - Purpose of factor analysis is to reduce dimensions. Reduction method of dimensionality is to use a few potential and unobservable random variables to describe covariance relationship of original variables.

- **Example 8.1.1** Linden did factor analysis for the scores of ten universal competitions from Olympics decathlon(十项全能) competition since world war II based on data from 139 athletes. The ten events are: 100m race(x_1), long jump(x_2)跳远, shot(x_3)铅球, high jump(x_4), 400m race(x_5), 110m hurdle race(x_6), discus(x_7)标枪, pole jump(x_8), javelin(x_9)铁饼, 1500m race(x_{10}). Factor analysis after standardization indicates that the scores can be contributed to athletes' **sprint speed**短跑速度, **explosive muscle**爆发性臂力, **explosive leg strength and endurance**爆发性腿力和耐力. Every aspect is called a factor. The following factor model describes the relation between the above ten scores and 4 factors:

$$x_i = \mu_i + a_{i1}f_1 + a_{i2}f_2 + a_{i3}f_3 + a_{i4}f_4 + \varepsilon_i, \quad i=1,2,\dots,10$$

where f_1, f_2, f_3, f_4 are factors, called **common factor** (公共因子). a_{ij} is called **loading**(载荷) of x_i on f_j , μ_i is mean of x_i , ε_i is the unexplainable part of x_i , called **specific factor** (特殊因子).

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- **Example 8.1.3** Boss interview 48 candidates and score their performance in the following 15 aspects :

x_1 : application form formation

x_9 : experience

x_2 : appearance

x_{10} : enthusiasm

x_3 : professional ability

x_{11} : ambition

x_4 : attractiveness

x_{12} : comprehensive ability

x_5 : self-confidence

x_{13} : potential

x_6 : smartness

x_{14} : social ability

x_7 : honesty

x_{15} : adaptability

x_8 : ability to sell yourself

- By means of factor analysis, 15 aspects can attribute to the following five factors given by **exposure ability**、**experience**、**attractive degree**、**professional ability** and **appearance** .



§ 8.2 Orthogonal Factor Model

- § 8.2.1 Mathematical Model
- § 8.2.2 Properties of Orthogonal Factor Model
- § 8.2.3 Statistical Significance of Factor Loading Matrix

§ 8.2.1 Mathematical Model

- Suppose p dimensional observable random vector $\mathbf{x} = (x_1, x_2, \dots, x_p)'$ and its mean and covariance matrix are $\boldsymbol{\mu} = (\mu_1, \mu_2, \dots, \mu_p)'$ and $\boldsymbol{\Sigma} = (\sigma_{ij})$. The model of factor analysis is

$$\begin{cases} x_1 = \mu_1 + a_{11}f_1 + a_{12}f_2 + \dots + a_{1m}f_m + \varepsilon_1 \\ x_2 = \mu_2 + a_{21}f_1 + a_{22}f_2 + \dots + a_{2m}f_m + \varepsilon_2 \\ \vdots \\ x_p = \mu_p + a_{p1}f_1 + a_{p2}f_2 + \dots + a_{pm}f_m + \varepsilon_p \end{cases} \quad (8.2.1)$$

where unobservable random variables $f_1, f_2, \dots, f_m$ are common factors. $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_p$ are special factors. Common factors are appeared in every original variable's expression. Namely, original variables have common factors. The above formula can be expressed in matrix

The formula (8.2.1) can be expressed in matrix

$$\mathbf{x} = \boldsymbol{\mu} + \mathbf{A}\mathbf{f} + \boldsymbol{\varepsilon} \quad (8.2.2)$$

where $\mathbf{f} = (f_1, f_2, \dots, f_m)'$ are common factor vectors, $\boldsymbol{\varepsilon} = (\varepsilon_1, \varepsilon_2, \dots, \varepsilon_p)'$ are special factor vectors. $\mathbf{A} = (a_{ij}) : p \times m$ is called **factor loading matrix**. Assume

$$\begin{cases} E(\mathbf{f}) = \mathbf{0} \\ E(\boldsymbol{\varepsilon}) = \mathbf{0} \\ V(\mathbf{f}) = \mathbf{I} \\ V(\boldsymbol{\varepsilon}) = \mathbf{D} = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_p^2) \\ \text{Cov}(\mathbf{f}, \boldsymbol{\varepsilon}) = E(\mathbf{f}\boldsymbol{\varepsilon}') = \mathbf{0} \end{cases} \quad (8.2.3)$$

- The above assumption and relation constitutes **orthogonal factor model**. Common factors and special factors are respectively mutually uncorrelated and have unit variance.

§ 8.2.2 Properties of Orthogonal Factor Model

- 1. decomposition of $\mathbf{x}$'s covariance matrix Σ
- 2. Model doesn't be influenced by unit.
- 3. Factor loading is not unique.

1. decomposition of $\mathbf{x}$'s covariance matrix Σ

$$\begin{aligned} V(\mathbf{x}) &= V(A\mathbf{f} + \boldsymbol{\varepsilon}) = \text{Cov}(A\mathbf{f} + \boldsymbol{\varepsilon}, A\mathbf{f} + \boldsymbol{\varepsilon}) \\ &= AV(\mathbf{f})A' + A\text{Cov}(\mathbf{f}, \boldsymbol{\varepsilon}) + \text{Cov}(\boldsymbol{\varepsilon}, \mathbf{f})A' + V(\boldsymbol{\varepsilon}) \\ &= AV(\mathbf{f})A' + V(\boldsymbol{\varepsilon}) \end{aligned}$$

$$\Sigma = AA' + D$$

If $\mathbf{x}$ is random vector with standardized component variables, then Σ is correlation matrix $R = (\rho_{ij})$. i.s.:

$$R = AA' + D$$

➤ **Example 8.2.1** Suppose the covariance matrix of random vector $\mathbf{x}=(x_1,x_2,x_3,x_4)'$ is

$$\Sigma = \begin{pmatrix} 9 & -11 & -5 & 20 \\ -11 & 27 & 17 & -42 \\ -5 & 17 & 52 & -5 \\ 20 & -42 & -5 & 86 \end{pmatrix}$$

Then, Σ is decomposed to

$$\Sigma = \mathbf{A}\mathbf{A}' + \mathbf{D}$$

where

$$\mathbf{A} = \begin{pmatrix} 2 & -1 \\ -4 & 3 \\ 1 & 7 \\ 9 & -2 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

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- Suppose $A = \Sigma^{\frac{1}{2}}, D = \mathbf{0}$, then the decomposition is

$$\Sigma = \Sigma^{\frac{1}{2}} \Sigma^{\frac{1}{2}} + \mathbf{0}$$

where $m=p$. It didn't reach the purpose of reducing dimensions. So, it is meaningless to do factor analysis.

- Out of the demand of dimensionality reduction, we usually hope that m is much smaller than p . Then the above decomposition of Σ usually can approximately set up, i.e.,

$$\Sigma \approx AA' + D$$

The higher degree of approximation is, the better factor model fitting is. Generally speaking, small m leads to bad approximate effect, namely unsatisfactory model fitting. Factor model with bad model fitting has no real meaning. So, m should not be too small.

2. Model doesn't be influenced by unit.

- Transform the unit of $\mathbf{x}$, usually $\mathbf{x}^* = \mathbf{C}\mathbf{x}$, where $\mathbf{C} = \text{diag}(c_1, c_2, \dots, c_p)$, $c_i > 0, i=1, 2, \dots, p$. So,

$$\mathbf{x}^* = \mathbf{C}\boldsymbol{\mu} + \mathbf{C}\mathbf{A}\mathbf{f} + \mathbf{C}\boldsymbol{\varepsilon}$$

Let $\boldsymbol{\mu}^* = \mathbf{C}\boldsymbol{\mu}$, $\mathbf{A}^* = \mathbf{C}\mathbf{A}$, $\boldsymbol{\varepsilon}^* = \mathbf{C}\boldsymbol{\varepsilon}$, then

$$\mathbf{x}^* = \boldsymbol{\mu}^* + \mathbf{A}^*\mathbf{f} + \boldsymbol{\varepsilon}^*$$

This model can satisfy the assumption of the above factor model, i.e.:

The assumptions of the orthogonal factor model.

$$\left\{ \begin{array}{l} E(\mathbf{f}) = \mathbf{0} \\ E(\boldsymbol{\varepsilon}^*) = \mathbf{0} \\ V(\mathbf{f}) = \mathbf{I} \\ V(\boldsymbol{\varepsilon}^*) = \mathbf{D}^* \\ \text{Cov}(\mathbf{f}, \boldsymbol{\varepsilon}^*) = \text{Cov}(\mathbf{f}, \boldsymbol{\varepsilon}) \mathbf{C}' = \mathbf{0} \end{array} \right.$$

where $\mathbf{D}^* = \text{diag}(\sigma_1^{*2}, \sigma_2^{*2}, \dots, \sigma_p^{*2})$, $\sigma_i^{*2} = c_i^2 \sigma_i^2$, $i = 1, 2, \dots, p$

Therefore, new model after unit transformation is still orthogonal factor model.

3. Factor loading is not unique.

- Suppose T is a $m \times m$ orthogonal matrix. Let $A^* = AT$, $f^* = T'f$, then model can be expressed as

$$x = \mu + A^*f^* + \varepsilon$$

Because

$$E(f^*) = T'E(f) = 0$$

$$V(f^*) = T'V(f)T = T'T = I$$

$$\text{Cov}(f^*, \varepsilon) = E(f^* \varepsilon') = T'E(f \varepsilon') = 0$$

so it still satisfies conditions of model. Σ can be decomposed to

$$\Sigma = A^*A^{*'} + D$$

- Therefore, factor loading matrix A is not unique. In real application, benefited from this fact, new factors will have better real meaning after factor rotation (§ 8.4), .

§ 8.2.3 Statistical Significance of Factor Loading Matrix

➤ 1. A 's elements a_{ij}

➤ 2. A sum of squares of row elements

$$h_i^2 = \sum_{j=1}^m a_{ij}^2$$

➤ 3. A sum of squares of column elements

$$g_j^2 = \sum_{i=1}^p a_{ij}^2$$

1. A 's elements a_{ij}

■ $x_i = \mu_i + a_{i1}f_1 + a_{i2}f_2 + \dots + a_{im}f_m + \varepsilon_i$

$$\text{Cov}(x_i, f_j) = \sum_{\alpha=1}^m a_{i\alpha} \text{Cov}(f_\alpha, f_j) + \text{Cov}(\varepsilon_i, f_j) = a_{ij}$$

a_{ij} is the **covariance** between x_i and f_j .

- If $\mathbf{x}$ is random vector with standardized component variables.
Correlation coefficient of x_i and f_j :

$$\rho(x_i, f_j) = \frac{\text{Cov}(x_i, f_j)}{\sqrt{V(x_i)V(f_j)}} = \text{Cov}(x_i, f_j) = a_{ij} \quad (8.2.9)$$

Now, a_{ij} is the **correlation coefficient** between x_i and f_j .

2. A sum of squares of row elements $h_i^2 = \sum_{j=1}^m a_{ij}^2$

$$x_i = \mu_i + a_{i1}f_1 + a_{i2}f_2 + \cdots + a_{im}f_m + \varepsilon_i$$

$$V(x_i) = a_{i1}^2 V(f_1) + a_{i2}^2 V(f_2) + \cdots + a_{im}^2 V(f_m) + V(\varepsilon_i)$$

$$= a_{i1}^2 + a_{i2}^2 + \cdots + a_{im}^2 + \sigma_i^2, \quad i = 1, 2, \cdots, p$$

Suppose $h_i^2 = \sum_{j=1}^m a_{ij}^2$, $i = 1, 2, \cdots, p$, then

$$\sigma_{ii} = h_i^2 + \sigma_i^2, \quad i = 1, 2, \cdots, p.$$

h_i^2 reflects the influence of common factors to x_i , which can be regarded as variance contribution of common factor $f_1, f_2, \cdots, f_m$ to x_i , called **communality variance** (共性方差). σ_i^2 is variance contribution of special factor ε_i to x_i , called **specific variance** (特殊方差).

When $\mathbf{x}$ is a random vector with standardized components, $\sigma_i = 1$. Then,

$$h_i^2 + \sigma_i^2 = 1, \quad i = 1, 2, \cdots, p.$$

3. A sum of squares of column elements $g_j^2 = \sum_{i=1}^p a_{ij}^2$

$$\begin{aligned}\sum_{i=1}^p V(x_i) &= \sum_{i=1}^p a_{i1}^2 + \cdots \sum_{i=1}^p a_{im}^2 + \sum_{i=1}^p \sigma_i^2 \\ &= g_1^2 + \cdots + g_m^2 + \sum_{i=1}^p \sigma_i^2.\end{aligned}$$

- Here, $g_j^2 = \sum_{i=1}^p a_{ij}^2$ reflects the influence of common factor f_j to $x_1, x_2, \dots, x_p$, which is a measure of the importance of the common factor f_j , and can be explained as total variance contribution of f_j to $x_1, x_2, \dots, x_p$.
- $\sum_{j=1}^m g_j^2$ represents the cumulative contribution of m common factors $f_1, \dots, f_m$ to the total variance $\sum_{i=1}^p V(x_i)$.



§ 8.3 Parameter Estimation

- § 8.3.1 principal component analysis method
- § 8.3.2 factorization method
- § 8.3.3 maximum likelihood method

§ 8.3.1 principal component analysis method

- Suppose the eigenvalues of sample covariance matrix $\mathbf{S}$ are $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \dots \geq \hat{\lambda}_p \geq 0$. Corresponding orthogonal unit eigenvectors are $\hat{\mathbf{t}}_1, \hat{\mathbf{t}}_2, \dots, \hat{\mathbf{t}}_p$. Choose a small factor number m in order to obtain a bigger percentage of accumulative contribution rate, where the accumulative contribution rate is $\sum_{i=1}^m \hat{\lambda}_i / \sum_{i=1}^p \hat{\lambda}_i$.

$\mathbf{S}$ is approximately decomposed to :

$$\begin{aligned} \mathbf{S} &= \hat{\lambda}_1 \hat{\mathbf{t}}_1 \hat{\mathbf{t}}_1' + \dots + \hat{\lambda}_m \hat{\mathbf{t}}_m \hat{\mathbf{t}}_m' + \hat{\lambda}_{m+1} \hat{\mathbf{t}}_{m+1} \hat{\mathbf{t}}_{m+1}' + \dots + \hat{\lambda}_p \hat{\mathbf{t}}_p \hat{\mathbf{t}}_p' \\ \mathbf{S} = \hat{\mathbf{A}} \hat{\mathbf{T}} \hat{\mathbf{A}}', \hat{\mathbf{T}} &= (\hat{\mathbf{t}}_1, \dots, \hat{\mathbf{t}}_p) \\ &\approx \hat{\lambda}_1 \hat{\mathbf{t}}_1 \hat{\mathbf{t}}_1' + \dots + \hat{\lambda}_m \hat{\mathbf{t}}_m \hat{\mathbf{t}}_m' + \hat{\mathbf{D}} = \hat{\mathbf{A}} \hat{\mathbf{A}}' + \hat{\mathbf{D}} \end{aligned} \quad (8.3.1)$$

where $\hat{\mathbf{A}} = (\sqrt{\hat{\lambda}_1} \hat{\mathbf{t}}_1, \dots, \sqrt{\hat{\lambda}_m} \hat{\mathbf{t}}_m) = (\hat{a}_{ij})$ is $p \times m$ matrix, $\hat{\mathbf{D}} = \text{diag}(\hat{\sigma}_1^2, \dots, \hat{\sigma}_p^2)$, $\hat{\sigma}_i^2 = s_{ii} - \sum_{j=1}^m \hat{a}_{ij}^2$, $i=1, 2, \dots, p$. $\hat{\mathbf{A}}$ and $\hat{\mathbf{D}}$ is one **principal component solution** (主成分解) of factor model.

§ 8.3.1 principal component analysis method

- $S - (\hat{A}\hat{A}' + \hat{D})$ is called as the residual matrix with diagonal elements being 0. When other non-diagonal elements are very small, we can think that the model with m factors fits the original data very well. For the principal component solution, we can prove that ([exercise 8.1](#))

The sum of elements square in residual matrix (8.3.2)

$$S - (\hat{A}\hat{A}' + \hat{D}) \leq \hat{\lambda}_{m+1}^2 + \cdots \hat{\lambda}_p^2$$

[Hint: the sum of elements square of $S - (\hat{A}\hat{A}' + \hat{D})$ is less than the sum of elements square of $S - \hat{A}\hat{A}'$ and $\leftarrow$

$$S - \hat{A}\hat{A}' = (\hat{t}_{m+1}, \dots, \hat{t}_p) \begin{bmatrix} \hat{\lambda}_{m+1} & & \\ & \ddots & \\ & & \hat{\lambda}_p \end{bmatrix} \begin{pmatrix} \hat{t}'_{m+1} \\ \vdots \\ \hat{t}'_p \end{pmatrix} \leftarrow$$

]

§ 8.3.1 principal component analysis method

- To principal component solution, when factor number is increasing, the former factor's estimation loading is unchanged. The total variance contribution of the j -th factor f_j to $\mathbf{x}$ is $\hat{\lambda}_j$, i.e

$$g_j^2 = \sum_{i=1}^p \hat{a}_{ij}^2 = \hat{\lambda}_j, \quad j = 1, 2, \dots, m$$

where the j - th column of $\hat{A}$, $\sqrt{\hat{\lambda}_j} \hat{t}_j = (\hat{a}_{1j}, \hat{a}_{2j}, \dots, \hat{a}_{1j})'$ from (8.3.1).

- **Example 8.3.1** In example 7.3.2, choose $m=1$ and $m=2$. Use principal component analysis to estimate factor loading and communality (共性方差) Table 8.3.1.

Table 8.3.1 principal component solution when $m=1$ and $m=2$

variables	$m=1$		$m=2$		
	factor loading	communality	factor loading		communality
	$f_1 \left(= \sqrt{\hat{\lambda}_1} \hat{t}_1 \right)$	$\hat{h}_i^2$	f_1	$f_2 \left(= \sqrt{\hat{\lambda}_2} \hat{t}_2 \right)$	$\hat{h}_i^2 (= \hat{a}_{i1}^2 + \hat{a}_{i2}^2)$
x_1^* : 100m	0.817	0.668	0.817	0.531	0.950
x_2^* : 200m	0.867	0.752	0.867	0.432	0.939
x_3^* : 400m	0.915	0.838	0.915	0.233	0.892
x_4^* : 800m	0.949	0.900	0.949	0.012	0.900
x_5^* : 1500m	0.959	0.920	0.959	-0.131	0.938
x_6^* : 5000m	0.938	0.879	0.938	-0.292	0.965
x_7^* : 10000m	0.944	0.891	0.944	-0.287	0.973
x_8^* : marathon	0.880	0.774	0.880	-0.411	0.943
explained total variance	0.828		0.828	0.938	$\frac{\sum_{j=1}^m g_j^2}{8}$
accumulative rate					

approximate relation of principal component solutions

$$\left\{ \begin{array}{l} x_1^*(100m) \approx 0.817f_1 + 0.531f_2 + \varepsilon_1 \\ x_2^*(200m) \approx 0.867f_1 + 0.432f_2 + \varepsilon_2 \\ x_3^*(400m) \approx 0.915f_1 + 0.233f_2 + \varepsilon_3 \\ x_4^*(800m) \approx 0.949f_1 + 0.012f_2 + \varepsilon_4 \\ x_5^*(1500m) \approx 0.959f_1 - 0.131f_2 + \varepsilon_5 \\ x_6^*(5000m) \approx 0.938f_1 - 0.292f_2 + \varepsilon_6 \\ x_7^*(10000m) \approx 0.944f_1 - 0.287f_2 + \varepsilon_7 \\ x_8^*(marathon) \approx 0.880f_1 - 0.411f_2 + \varepsilon_8 \end{array} \right.$$

- Factor explanation of principal component solution is exactly the same as principal component explanation. f_1 represents total strength in running events, called **strength factor (强弱因子)**. F_2 reflects **comparison of speed and endurance**.

Table 8.3.2 residual matrix of principal component solution when $m=2$

$$\hat{R} - \hat{A}\hat{A}' - \hat{D} = \begin{bmatrix} 0.000 & & & & & & & \\ -0.016 & 0.000 & & & & & & \\ -0.030 & -0.044 & 0.000 & & & & & \\ -0.025 & -0.021 & -0.001 & 0.000 & & & & \\ -0.014 & -0.000 & -0.012 & 0.009 & 0.000 & & & \\ 0.008 & 0.009 & -0.012 & -0.023 & -0.010 & 0.000 & & \\ 0.014 & 0.002 & -0.010 & -0.023 & -0.008 & 0.006 & 0.000 & \\ 0.019 & 0.011 & -0.005 & -0.024 & -0.033 & -0.013 & -0.006 & 0.000 \end{bmatrix}$$

$$\hat{D} = \text{diag}(\hat{\sigma}_1^2, \dots, \hat{\sigma}_p^2), \quad \hat{\sigma}_i^2 = s_{ii} - \sum_{j=1}^m \hat{a}_{ij}^2$$

§ 8.3.2 factorization method

- Assume $\mathbf{x}$ is a random vector with standardized component variables. If $\mathbf{x}$ satisfies orthogonal factor model. Then

$$\mathbf{R} = \mathbf{A}\mathbf{A}' + \mathbf{D}$$

where $\mathbf{R}$ is the correlation matrix of $\mathbf{x}$. Suppose

$$\mathbf{R}^* = \mathbf{R} - \mathbf{D} = \mathbf{A}\mathbf{A}'$$

then $\mathbf{R}^*$ is the **reduced correlation matrix** of $\mathbf{x}$ (约相关矩阵).

- The diagonal elements of $\mathbf{R}^*$ are h_i^2 not 1. Non-diagonal elements are exactly the same as in $\mathbf{R}$. And $\mathbf{R}^*$ is a non-negative matrix.

- Suppose $\hat{\sigma}_i^2$ is a proper initial estimation of special variance σ_i^2 . Then **reduced correlation matrix (约相关矩阵)** can be estimated as:

$$\hat{\mathbf{R}}^* = \hat{\mathbf{R}} - \hat{\mathbf{D}} = \begin{pmatrix} \hat{h}_1^2 & r_{12} & \cdots & r_{1p} \\ r_{21} & \hat{h}_2^2 & \cdots & r_{2p} \\ \vdots & \vdots & & \vdots \\ r_{p1} & r_{p2} & \cdots & \hat{h}_p^2 \end{pmatrix}$$

where $\hat{\mathbf{R}} = (r_{ij})$, $\hat{\mathbf{D}} = \text{diag}(\hat{\sigma}_1^2, \hat{\sigma}_2^2, \dots, \hat{\sigma}_p^2)$, $\hat{h}_i^2 = 1 - \hat{\sigma}_i^2$ is initial estimation of h_i^2 . Suppose the first m eigenvalues of $\hat{\mathbf{R}}^*$ are $\hat{\lambda}_1^* \geq \hat{\lambda}_2^* \geq \dots \geq \hat{\lambda}_m^* > 0$ and corresponding orthogonal unit eigenvectors are $\hat{\mathbf{t}}_1^*, \hat{\mathbf{t}}_2^*, \dots, \hat{\mathbf{t}}_m^*$. Then, **principal factor solution** of $\mathbf{A}$:

$$\hat{\mathbf{A}} = \left(\sqrt{\hat{\lambda}_1^*} \hat{\mathbf{t}}_1^*, \sqrt{\hat{\lambda}_2^*} \hat{\mathbf{t}}_2^*, \dots, \sqrt{\hat{\lambda}_m^*} \hat{\mathbf{t}}_m^* \right)$$

which is principal component solution (主因子解) based on $\hat{\mathbf{R}}^*$.

- Therefore, we can re-estimate special variance. Final estimation of σ_i^2 is

$$\hat{\sigma}_i^2 = 1 - \hat{h}_i^2 = 1 - \sum_{j=1}^m \hat{a}_{ij}^2, \quad i = 1, 2, \dots, p$$

We can use iteration method to obtain solution with higher fitting degree.

Use $\hat{\sigma}_i^2$ as the initial estimation of special variance. Repeat the above steps until the solution is stable.

- The multiple correlation coefficient between y and x is defined as

$$\rho_{y \cdot x} = \rho_{y \cdot 1, 2, \dots, p} = \max_{l \neq 0} \rho(y, l'x) = \left(\frac{\sigma'_{xy} \Sigma_{xx}^{-1} \sigma_{xy}}{\sigma_{yy}} \right)^{1/2}$$

- If $x_1, x_2, \dots, x_p$ are not correlated, then $R_{xx} = I$. This leads to

$$\rho_{y \cdot x}^2 = \rho'_{xy} R_{xx}^{-1} \rho_{xy} = \rho'_{xy} \rho_{xy} = \rho^2(y, x_1) + \dots + \rho^2(y, x_p)$$

- It is noticed that $h_i^2 = \sum_{j=1}^m a_{ij}^2$ if the components x_j 's in the original vector x are standardized. This implies that h_i^2 is the square of multiple correlation coefficient between x_i and m common factors.

a_{ij} is the **correlation coefficient** between x_i and f_j

Common initial estimation method of special (communality) variance

- (1) Let communality variance's estimation $\hat{h}_i^2$ be the square of multiple correlation coefficient between x_i and other $p-1$ variables, then $\hat{\sigma}_i^2 = 1 - \hat{h}_i^2$. The above method is a common practice to set the initial estimation of σ_i^2 .
- (2) Choose $\hat{h}_i^2 = \max_{j \neq i} |r_{ij}|^2$.
- (3) Choose $\hat{h}_i^2 = 1$ such that $\hat{\sigma}_i^2 = 0$. We can obtain a principal component solution of $\hat{A}$.

- **Example 8.3.2** In example 7.3.2, choose $m=2$. In order to obtain principal factor solution, choose squares of multiple correlation coefficient between x_i and other 7 variables as the initial estimation of h_i^2 . Calculate

$$\hat{h}_1^2 = 0.877, \quad \hat{h}_2^2 = 0.888, \quad \hat{h}_3^2 = 0.845, \quad \hat{h}_4^2 = 0.884$$

$$\hat{h}_5^2 = 0.927, \quad \hat{h}_6^2 = 0.955, \quad \hat{h}_7^2 = 0.967, \quad \hat{h}_8^2 = 0.905$$

So, reduced correlation matrix is

$$\hat{R}^* = \begin{pmatrix} 0.877 & & & & & & & \\ 0.923 & 0.888 & & & & & & \\ 0.841 & 0.851 & 0.845 & & & & & \\ 0.756 & 0.807 & 0.870 & 0.884 & & & & \\ 0.700 & 0.775 & 0.835 & 0.918 & 0.927 & & & \\ 0.619 & 0.695 & 0.779 & 0.864 & 0.928 & 0.955 & & \\ 0.633 & 0.697 & 0.787 & 0.869 & 0.935 & 0.975 & 0.967 & \\ 0.520 & 0.596 & 0.705 & 0.806 & 0.866 & 0.932 & 0.943 & 0.905 \end{pmatrix}$$



$\hat{R}^*$'s eigenvalues are

$$\hat{\lambda}_1^* = 6.530, \quad \hat{\lambda}_2^* = 0.779, \quad \hat{\lambda}_3^* = 0.051, \quad \hat{\lambda}_4^* = 0.006$$

$$\hat{\lambda}_5^* = -0.014, \quad \hat{\lambda}_6^* = -0.015, \quad \hat{\lambda}_7^* = -0.036, \quad \hat{\lambda}_8^* = -0.053$$

Since $\hat{\lambda}_3^*$ eigenvalues are almost 0. So, choose $m=2$.

Result is listed in Table 8.3.2.

Table 8.3.2

principal factor solution when $m=2$

variables	Factor loading $\hat{A} = \left(\sqrt{\hat{\lambda}_1^* \hat{t}_1^*}, \sqrt{\hat{\lambda}_2^* \hat{t}_2^*} \right)$		communality
	f_1	f_2	$\hat{h}_i^2$
x_1^* : 100m	0.807	0.496	0.897
x_2^* : 200m	0.858	0.412	0.906
x_3^* : 400m	0.890	0.216	0.856
x_4^* : 800m	0.939	0.024	0.881
x_5^* : 1500m	0.956	-0.114	0.926
x_6^* : 5000m	0.938	-0.282	0.960
x_7^* : 10000m	0.946	-0.281	0.974
x_8^* : marathon	0.874	-0.378	0.907
Accumulative proportion of explained total variance	0.816	0.914	$\frac{\sum_{j=1}^m g_j^2}{8}$

$$0.923 - 0.858 \times 0.807 - 0.412 \times 0.496 = 0.0262$$

Table 8.3.2' residual matrix of principal factor solution when $m=2$

$$\hat{R} - \hat{A}\hat{A}' - \hat{D}$$

$$\hat{R}^* - \hat{A}\hat{A}' = \begin{bmatrix} 0.000 & & & & & & & \\ 0.026 & 0.000 & & & & & & \\ 0.008 & -0.010 & 0.000 & & & & & \\ -0.013 & -0.008 & 0.021 & 0.000 & & & & \\ -0.014 & 0.002 & 0.000 & 0.024 & 0.000 & & & \\ 0.002 & 0.007 & -0.005 & -0.010 & -0.001 & 0.000 & & \\ 0.009 & 0.001 & -0.003 & -0.012 & -0.002 & 0.007 & 0.000 & \\ 0.002 & 0.002 & 0.000 & -0.005 & -0.013 & 0.005 & 0.010 & 0.000 \end{bmatrix}$$

- Comparing the residual matrix of principal component decomposition and principal factor solution, it can be seen that the main factor solution is better fitted a little.

§ 8.3.3 maximum likelihood method

- Suppose common factor $\mathbf{f} \sim N_m(\mathbf{0}, \mathbf{I})$ and special factor $\boldsymbol{\varepsilon} \sim N_p(\mathbf{0}, \mathbf{D})$. They are independent. Then, original variable $\mathbf{x} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. From sample $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$, we obtain likelihood function is $L(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Since $\boldsymbol{\Sigma} = \mathbf{A}\mathbf{A}' + \mathbf{D}$, likelihood function can be clearly expressed as $L(\boldsymbol{\mu}, \mathbf{A}, \mathbf{D})$. Maximum likelihood estimation of $(\boldsymbol{\mu}, \mathbf{A}, \mathbf{D})$ is $(\hat{\boldsymbol{\mu}}, \hat{\mathbf{A}}, \hat{\mathbf{D}})$.

$$L(\hat{\boldsymbol{\mu}}, \hat{\mathbf{A}}, \hat{\mathbf{D}}) = \max L(\boldsymbol{\mu}, \mathbf{A}, \mathbf{D})$$

- $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$. $\hat{\mathbf{A}}$ and $\hat{\mathbf{D}}$ satisfy the following equation set:

$$\begin{cases} \hat{\boldsymbol{\Sigma}}\hat{\mathbf{D}}^{-1}\hat{\mathbf{A}} = \hat{\mathbf{A}}(\mathbf{I}_m + \hat{\mathbf{A}}'\hat{\mathbf{D}}^{-1}\hat{\mathbf{A}}) \\ \hat{\mathbf{D}} = \text{diag}(\hat{\boldsymbol{\Sigma}} - \hat{\mathbf{A}}\hat{\mathbf{A}}') \end{cases}$$

The details of derivation can be found in Page 264-265 of the textbook (Multivariate Analysis by K.V. MARDIA, et.al. Academic Press)

where $\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})'$. Since the solutions of $\mathbf{A}$ are not unique, we

can add uniqueness condition to obtain unique solution:

$\mathbf{A}'\mathbf{D}^{-1}\mathbf{A}$ is diagonal matrix

$\hat{\mathbf{A}}$ and $\hat{\mathbf{D}}$ generally can be solved by iteration.

- For maximum likelihood solution, when factor number increases, the original factors' estimated loading and contribution to $\mathbf{x}$ will change. It is different from principal component solution and principal factor solution.
- **Example 8.3.3** In example 7.3.2, choose $m=2$. Calculation results of maximum likelihood solution are listed in Table 8.3.3.

Table 8.3.3 $m=2$, maximum likelihood solution

variables	Factor loading		communality
	f_1	f_2	$\hat{h}_i^2$
x_1^* : 100m	0.731	-0.620	0.919
x_2^* : 200m	0.792	-0.545	0.924
x_3^* : 400m	0.855	-0.343	0.849
x_4^* : 800m	0.916	-0.161	0.865
x_5^* : 1500m	0.958	-0.026	0.918
x_6^* : 5000m	0.972	0.144	0.966
x_7^* : 10000m	0.981	-0.143	0.982
x_8^* : marathon	0.923	-0.249	0.914
Accumulative proportion of explained total variance	0.801	0.917	

Table 8.3.3 residual matrix of principal factor solution when $m=2$

$$\hat{R} - \hat{A}\hat{A}' - \hat{D} = \begin{bmatrix} 0.000 & & & & & & & \\ 0.026 & 0.000 & & & & & & \\ 0.008 & -0.010 & 0.000 & & & & & \\ -0.013 & -0.008 & 0.021 & 0.000 & & & & \\ -0.014 & 0.002 & 0.000 & 0.024 & 0.000 & & & \\ 0.002 & 0.007 & -0.005 & -0.010 & -0.001 & 0.000 & & \\ 0.009 & 0.001 & -0.003 & -0.012 & -0.002 & 0.007 & 0.000 & \\ 0.002 & 0.002 & 0.000 & -0.005 & -0.013 & 0.005 & 0.010 & 0.000 \end{bmatrix}$$

- The fitting of maximum likelihood method seems to be slightly better than that of the principal factor method.



Assignment: Assignment has been
published on BB



§ 8.4 Factor Rotation

- Explanation of factor is more subjective. We usually reduce subjectivity by rotating common factors.
- To some extent, whether common factors are easy to explain depends on element structure of factor loading matrix A .
- If all the elements of loading matrix A approach to 0 or ± 1 , common factors are easy to explain. Otherwise, if A 's elements are neither too big or too little in majority, then common factors are not easy to explain. By using factor rotation, absolute values of elements on each columns of loading matrix have big difference.

- Factor rotation methods include **orthogonal rotation** and **oblique rotation**. Only orthogonal rotation will be discussed.
- Orthogonal rotation of common factors is equivalent to make an orthogonal transformation for loading matrix A so that the new loading matrix $A^* = AT$ has brighter meaning. After rotation, the geometric meaning $f^* = T'f$ is the orthogonal rotation in m dimensional space.

$$X = \mu + Af + \varepsilon = \mu + \underbrace{AT}_{A^*} \underbrace{T'f}_{f^*} + \varepsilon$$

- Factor rotation doesn't change communality **variance** (共性方差).

It is noticed that

$$A^*A^{*'} = ATT'A' = AA'$$

Let $A = (a_1, a_2, \dots, a_p)'$ and $A^* = (a_1^*, a_2^*, \dots, a_p^*)'$, then

$$h_i^{*2} = a_i^{*'} a_i^* = a_i' a_i = h_i^2, \quad i = 1, 2, \dots, p$$

- Factor rotation doesn't change residual matrix since

$$\Sigma = AA' + D \tag{8.3.2}$$

$$\Sigma - AA' = \Sigma - A^*A^{*'} = D$$

- Orthogonal rotation has many methods. They are based on the choice of orthogonal matrix T . The most common one is **varimax rotation** (最大方差旋转法). Only this method will be introduced in this section.
- Suppose $A^* = AT = (a_{ij}^*)$, $d_{ij} = a_{ij}^* / h_i$, $\bar{d}_j = \frac{1}{p} \sum_{i=1}^p d_{ij}^2$. Here, a_{ij}^* / h_i is to eliminate the different contribution of common factors to the variance of original variables. Then relative variance of squares of j-th column elements of A^* can be defined as

$$V_j = \frac{1}{p} \sum_{i=1}^p (d_{ij}^2 - \bar{d}_j)^2 \quad (8.4.3)$$

- Then the varimax rotation is to find an orthogonal matrix T to maximize sum of relative variance of squares of all column elements in

$$V = V_1 + V_2 + \cdots + V_m \quad (8.4.4)$$

When $m=2$, suppose factor loading matrix is $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ \vdots & \vdots \\ a_{p1} & a_{p2} \end{pmatrix}$

- Choose orthogonal transformation matrix T to implement factor rotation. T can be expressed by

$$T = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix},$$

where θ is anti-clockwise rotation angle of factor axis on coordinate planes. Then,

$$\begin{aligned} A^* &= AT \\ &= \begin{bmatrix} a_{11} \cos \theta + a_{12} \sin \theta & -a_{11} \sin \theta + a_{12} \cos \theta \\ a_{21} \cos \theta + a_{22} \sin \theta & -a_{21} \sin \theta + a_{22} \cos \theta \\ \vdots & \vdots \\ a_{p1} \cos \theta + a_{p2} \sin \theta & -a_{p1} \sin \theta + a_{p2} \cos \theta \end{bmatrix} = \begin{bmatrix} a_{11}^* & a_{12}^* \\ a_{21}^* & a_{22}^* \\ \vdots & \vdots \\ a_{p1}^* & a_{p2}^* \end{bmatrix} \end{aligned}$$

$$d_{ij} = a_{ij}^*/h_i, i=1,2,\dots,p, j=1,2, \quad \bar{d}_j = \frac{1}{p} \sum_{i=1}^p d_{ij}^2, j=1,2$$

We need to maximize $V \equiv V(\theta) = V_1 + V_2 + \dots + V_p$ to find θ .

Example 8.4.1. Perform factor analysis on ten variables starting from R and select two factors, f_1 and f_2 . In Figure 8.4.1, there are coordinate points for these eight variables, with the horizontal axis f_1 and vertical axis f_2 representing the loads of the variables on factors f_1 and f_2 , respectively. The meaning of the f_1^* and f_2^* axes after coordinate rotation is similar. The factor load after rotation clearly has a simple structure, and its rotation angle can be subjectively determined by visual inspection in this figure.

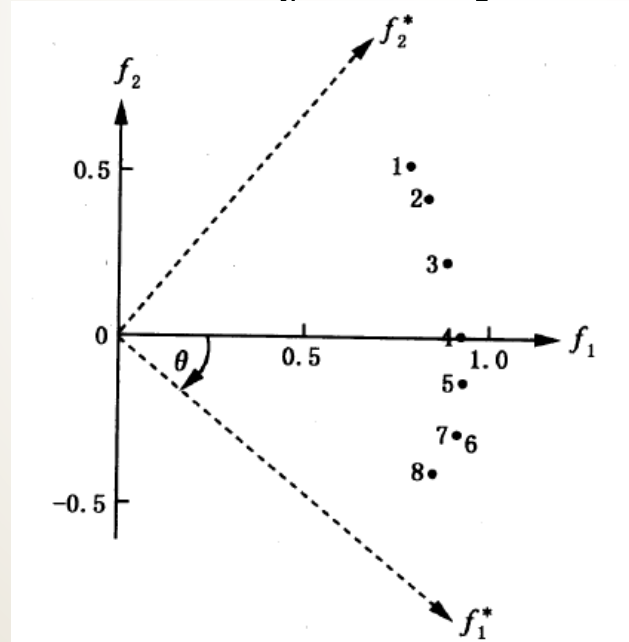


Fig 8.4.1 Factor load diagram with simple structure after rotation

当 $m>2$ 时，我们可以逐次对每两个因子进行上述旋转。对因子 f_l 和 f_k 进行旋转，就是对 A 的第 l 和 k 两列进行正交变换，使得这两列元素平方的相对方差之和达到最大，其余列不变。其正交变换矩阵为

$$T_{lk} = \begin{bmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & & & & \\ & & & \cos \theta & & -\sin \theta & \\ & & & & 1 & & \\ & & & & & \ddots & \\ & & & & & & 1 \\ & & \sin \theta & & & \cos \theta & \\ & & & & & & 1 & \ddots & \\ & & & & & & & & 1 \end{bmatrix}$$

l k

m 个因子的两两配对旋转共需进行 $\binom{m}{2} = \frac{1}{2} m(m-1)$ 次，称其为完成第一次旋转，并记第一轮旋转后得到的因子载荷矩阵为 $A^{(1)}$ 。

如此继续旋转下去，记第s论旋转后的因子载荷矩阵为 $A^{(s)}$,得到一系列因子载荷矩阵为

$$A^{(1)}, A^{(2)}, \dots, A^{(s)}, \dots$$

记 $V^{(s)}$ 为各列元素平方的相对方差之和，则必然有

$$V^{(1)} \leq V^{(2)} \leq \dots \leq V^{(s)} \dots$$

这是一个有界的单调上升数列,必收敛。当 $V^{(s)}$ 的值变化不大时即可停止旋转.

- 
- **Example 8.4.1.** In examples 8.3.1 to 8.3.3, **varimax rotation** is used, respectively, and factor loading matrix after rotation is listed in Table 8.4.1.

Table 8.4.1

variables	factor loading estimation after rotation					
	Principal components		Principal factors		Maximum likelihood	
	f_1^*	f_2^*	f_1^*	f_2^*	f_1^*	f_2^*
x_1^* : 100m	0.274	0.935	0.287	0.903	0.288	0.914
x_2^* : 200m	0.376	0.893	0.381	0.872	0.379	0.883
x_3^* : 400m	0.543	0.773	0.541	0.751	0.541	0.746
x_4^* : 800m	0.712	0.627	0.695	0.631	0.689	0.624
x_5^* : 1500m	0.813	0.525	0.799	0.537	0.797	0.532
x_6^* : 5000m	0.902	0.389	0.895	0.399	0.899	0.397
x_7^* : 10000m	0.903	0.397	0.900	0.405	0.906	0.402
x_8^* : marathon	0.936	0.261	0.909	0.284	0.914	0.281
Accumulative proportion of explained total variance	0.523	0.938	0.510	0.914	0.512	0.917

- 
- 
- By using three estimation methods of factor loading, we obtain approximate results after factor rotation. Loading of f_1^* is increasing successively, while loading of f_2^* is decreasing. f_1^* is called **Endurance Factor**. f_2^* is called **Speed Factor**.
 - (Principal component solutions) In figure 8.4.1, the horizontal axis f_1 represents the loading of the original variable on factor f_1 , and the vertical axis f_2 represents the loading of the original variable on factor f_2 . Factor loading pairs $(\hat{a}_{i1}, \hat{a}_{i2})$ of principal component decomposition are represented by points in figure 8.4.1. After varimax rotation, factors rotate clockwise by $\theta=40.6$. The coordinates of point i in the new coordinate system are the paired factor loadings $(\hat{a}_{i1}^*, \hat{a}_{i2}^*)$ after rotation. We can easily figure out factors' real meaning after rotation from the figure.

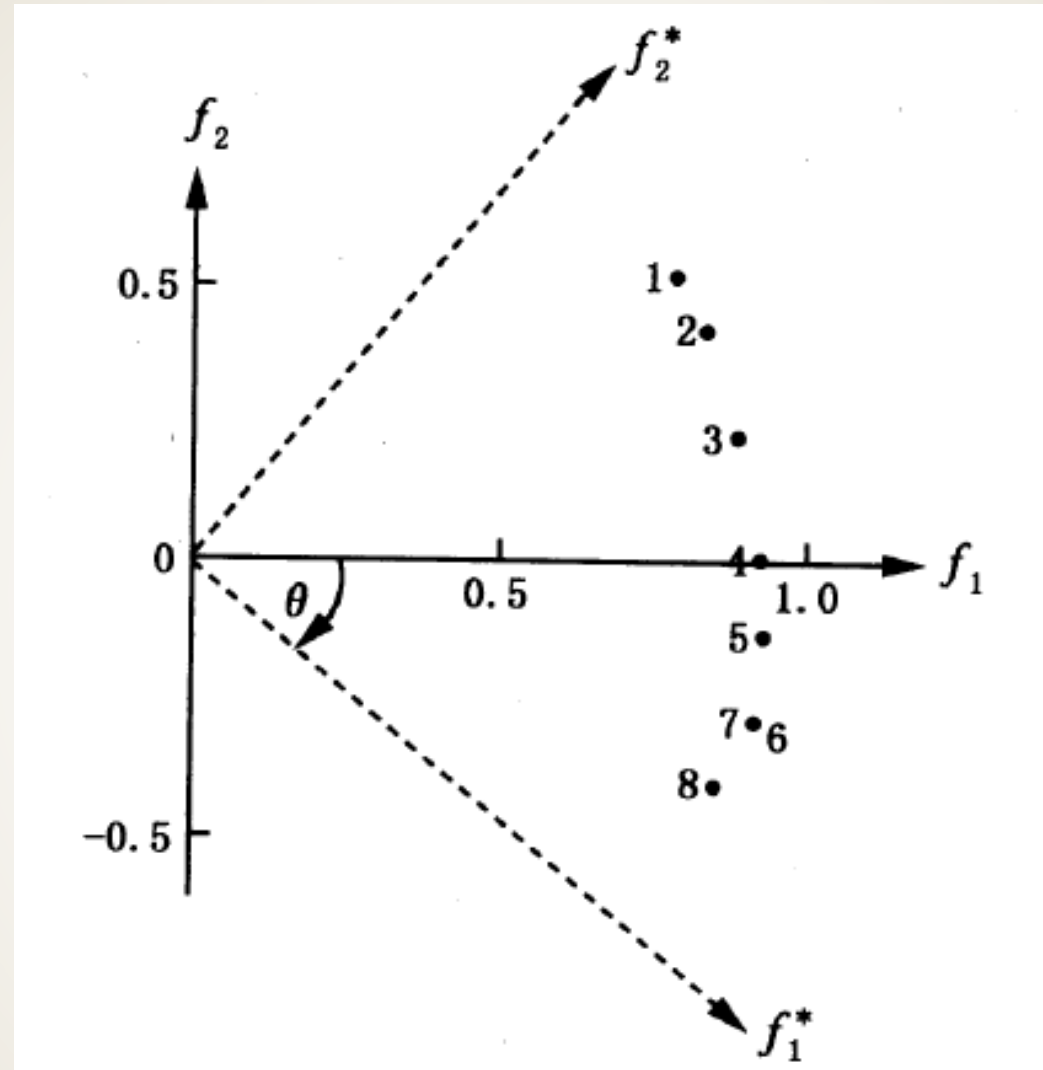


Figure 8.4.1 Factor Rotation

► **Example 8.4.2** 604 public companies in Shanghai Stock Market have ten main financial indexes in their financial statement of year 2001 :

x_1 : main business income (元) x_6 : BVPS(元),每股净资产

x_2 : main business profit (元) x_7 : ROE(%), 净资产收益率

x_3 : total profit(元) x_8 : ROA(%), 总资产收益率

x_4 : net profit(元) x_9 : total assets 资产总计(元)

x_5 : EPS(元), 每股收益 x_{10} : capital stock, 股本

The sample correlation matrix of the above 10 indicators is in Table 8.4.2.

Table 8.4.2

Sample Correlation Matrix

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}
x_1	1.000									
x_2	0.723	1.000								
x_3	0.427	0.743	1.000							
x_4	0.407	0.697	0.982	1.000						
x_5	0.171	0.325	0.539	0.559	1.000					
x_6	0.149	0.228	0.284	0.274	0.585	1.000				
x_7	0.096	0.177	0.362	0.402	0.776	0.218	1.000			
x_8	0.066	0.204	0.455	0.500	0.849	0.290	0.833	1.000		
x_9	0.748	0.768	0.574	0.567	0.125	0.138	0.067	0.058	1.000	
x_{10}	0.622	0.619	0.485	0.500	0.002	-0.066	0.033	0.051	0.861	1.000

- Starting from the correlation matrix, the first four eigenvalues of the correlation matrix of the principal component method are selected as

$$\hat{\lambda}_1 = 4.879, \quad \hat{\lambda}_2 = 2.574, \quad \hat{\lambda}_3 = 0.929 \quad \hat{\lambda}_4 = 0.713.$$

Accumulative contribution rate of three common factors is 83.82%.

Thus, we choose $m=3$ and list the result in Table 8.4.3.

Table 8.4.3

 $m=3$, principal component solution

variables	Factor loading			communality
	f_1	f_2	f_3	$\hat{h}_i^2$
x_1^* : main business income	0.659	-0.472	0.121	0.672
x_2^* : main business profit	0.835	-0.346	0.097	0.826
x_3^* : total profit	0.886	0.003	-0.037	0.786
x_4^* : net profit	0.888	0.037	-0.082	0.796
x_5^* : EPS	0.666	0.692	0.109	0.934
x_6^* : BVPS	0.391	0.367	0.814	0.951
x_7^* : ROE	0.527	0.670	-0.325	0.832
x_8^* : ROA	0.581	0.703	-0.260	0.899
x_9^* : total assets	0.747	-0.564	0.019	0.877
x_{10}^* : capital stock	0.636	-0.596	-0.219	0.808
Accumulative proportion of explained total variance	0.488	0.745	0.838	

Table 8.4.4

Factor loading estimation after rotation

variables	Factor Loading			Communality Variance $\hat{h}_i^2$
	f_1^*	f_2^*	f_3^*	
x_1^* : main business income	0.809	-0.029	0.129	0.672
x_2^* : main business profit	0.874	0.171	0.182	0.826
x_3^* : total profit	0.706	0.509	0.167	0.786
x_4^* : net profit	0.688	0.552	0.135	0.796
x_5^* : EPS (每股收益)	0.115	0.849	0.447	0.934
x_6^* : BVPS (每股净资产)	0.082	0.199	0.951	0.951
x_7^* : ROE (净资产收益率)	0.022	0.912	0.004	0.832
x_8^* : ROA (总资产收益率)	0.045	0.943	0.087	0.899
x_9^* : total assets (总资产)	0.936	-0.012	0.028	0.877
x_{10}^* : capital stock (股本)	0.869	-0.013	-0.228	0.808
Accumulative proportion of explained total variance	0.404	0.712	0.838	

f_1^* : scale factor, f_2^* : profit factor, f_3^* : value per share factor,

Remarks :

- $x_1^*, x_2^*, x_3^*, x_4^*, x_9^*, x_{10}^*$ have large positive loadings on factor f_1^* , and $x_5^*, x_6^*, x_7^*, x_8^*$ have small loadings on factor f_1^* . Therefore, this factor can be called as the scale factor (规模因子) of stock.
- x_5^*, x_7^*, x_8^* have large positive loadings on factor f_2^* , and x_3^*, x_4^* have medium positive loadings on factor f_2^* , the remaining variables have small loadings on f_2^* . Therefore, this factor can be called as the profit factor (盈利因子) of stock.
- x_6^* has large positive loadings on factor f_3^* , x_5^* has medium positive loadings on f_3^* . The remaining variables have small loadings on f_3^* . Therefore, this factor can be called as the value factor (价值因子) of stock.

§ 8.5 Factor Score

► Factor Model:

$$\boldsymbol{x} = \boldsymbol{\mu} + A\boldsymbol{f} + \boldsymbol{\varepsilon} \quad (8.2.1)$$

- Our aim: Provide scores of m common factors of each sample $\boldsymbol{x}_j$, where $\boldsymbol{x}_j$ is the j -th observation for $\boldsymbol{x}$.
- In this way, m common factor scores can be used to replace the p original variables, so as to achieve the purpose of comparative analysis of each sample after dimension reduction
- To estimate the value of unobservable random variable $f_1, f_2, \dots, f_m$.



§ 8.5 Factor Score

- § 8.5.1 weighted least square method
- § 8.5.2 regression method

§ 8.5.1 weighted least square method

- Use similar idea as weighted least square estimation in the analysis of regression, i.e.: minimizing weighted residual sum of squares:

$$\sum_{i=1}^p [(x_i - \mu_i) - (a_{i1}\hat{f}_1 + a_{i2}\hat{f}_2 + \cdots + a_{im}\hat{f}_m)]^2 / \sigma_i^2$$

Matrix form: $(X - \mu - A\hat{f})' D^{-1} (X - \mu - A\hat{f})$ (8.5.2)

By maximizing (8.5.2), we obtain the estimation of f as

$$\hat{f} = (A'D^{-1}A)^{-1} A'D^{-1}(x - \mu) \quad (8.5.3)$$

- In real application, replace μ , A and D by $\bar{x}$, $\hat{A}$ and $\hat{D}$ and substitute data of sample x_j in (8.5.3). Then the relevant factor score:

$$\hat{f}_j = (\hat{A}'\hat{D}^{-1}\hat{A})^{-1} \hat{A}'\hat{D}^{-1}(x_j - \bar{x}) \quad (8.5.4)$$

- For factor scores obtained through WLS method, assume $\mathbf{f}$ and $\boldsymbol{\varepsilon}$ are mutually independent. Then conditional expectation of factor score of $\hat{\mathbf{f}}$:

$$E(\hat{\mathbf{f}} | \mathbf{f}) = (\mathbf{A}'\mathbf{D}^{-1}\mathbf{A})^{-1} \mathbf{A}'\mathbf{D}^{-1} E(\mathbf{A}\mathbf{f} + \boldsymbol{\varepsilon} | \mathbf{f}) = \mathbf{f}.$$

- Notice that

$$\begin{aligned} \hat{\mathbf{f}} - \mathbf{f} &= (\mathbf{A}'\mathbf{D}^{-1}\mathbf{A})^{-1} \mathbf{A}'\mathbf{D}^{-1} (\mathbf{A}\mathbf{f} + \boldsymbol{\varepsilon}) - \mathbf{f} \\ &= (\mathbf{A}'\mathbf{D}^{-1}\mathbf{A})^{-1} \mathbf{A}'\mathbf{D}^{-1} \boldsymbol{\varepsilon}, \end{aligned}$$

we obtain that **average forecast error of estimation** $\hat{\mathbf{f}}$:

$$\begin{aligned} &E[(\hat{\mathbf{f}} - \mathbf{f})(\hat{\mathbf{f}} - \mathbf{f})'] \\ &= (\mathbf{A}'\mathbf{D}^{-1}\mathbf{A})^{-1} \mathbf{A}'\mathbf{D}^{-1} E(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}') \mathbf{D}^{-1} \mathbf{A} (\mathbf{A}'\mathbf{D}^{-1}\mathbf{A})^{-1} \\ &= (\mathbf{A}'\mathbf{D}^{-1}\mathbf{A})^{-1} \mathbf{A}'\mathbf{D}^{-1} \mathbf{D}\mathbf{D}^{-1} \mathbf{A} (\mathbf{A}'\mathbf{D}^{-1}\mathbf{A})^{-1} \\ &= (\mathbf{A}'\mathbf{D}^{-1}\mathbf{A})^{-1} \end{aligned}$$

(8.5.14)

§ 8.5.2 regression method

- In orthogonal factor model, suppose $\begin{pmatrix} f \\ x \end{pmatrix}$ obeys $(m+p)$ normal distribution

$$E\begin{pmatrix} f \\ x \end{pmatrix} = \begin{pmatrix} E(f) \\ E(x) \end{pmatrix} = \begin{pmatrix} 0 \\ \mu \end{pmatrix} \quad V\begin{pmatrix} f \\ x \end{pmatrix} = E\left[\begin{pmatrix} f \\ x - \mu \end{pmatrix} (f', (x - \mu)')\right] = \begin{pmatrix} I & A' \\ A & \Sigma \end{pmatrix}$$

- Using regression forecast method, estimation of $f = (f_1, f_2, \dots, f_m)'$ will be

$$\tilde{f} = E(f | x) = A' \Sigma^{-1} (x - \mu) \quad (8.5.8)$$

by using the formula of conditional expectation

$$\mu_{1.2} = \mu_1 + \Sigma_{12} \Sigma_{22}^{-1} (x_2 - \mu_2). \quad (3.2.6)$$

It is known $\Sigma = AA' + D$, we rewrite (8.5.8) as

$$\tilde{f} = A'(AA' + D)^{-1} (x - \mu) \quad (8.5.9)$$

or
$$\tilde{f} = (I + A'D^{-1}A)^{-1} A'D^{-1} (x - \mu) \quad (8.5.10)$$

by using $A'(AA' + D)^{-1} = (I + A'D^{-1}A)^{-1} A'D^{-1}$.

Exercise 8.1. Prove that

$$A^T (AA^T + D)^{-1} = (I + A^T D^{-1} A)^{-1} A^T D^{-1}$$

Hints: by using Woodbury matrix identity

$$(AA^T + D)^{-1} = D^{-1} - D^{-1} A (I + A^T D^{-1} A)^{-1} A^T D^{-1},$$

we obtain

$$\begin{aligned} & A^T (AA^T + D)^{-1} \\ &= \{I - A^T D^{-1} A (I + A^T D^{-1} A)^{-1}\} A^T D^{-1} \\ &= [(I + A^T D^{-1} A)(I + A^T D^{-1} A)^{-1} - A^T D^{-1} A (I + A^T D^{-1} A)^{-1}] A^T D^{-1} \\ &= (I + A^T D^{-1} A)^{-1} A^T D^{-1}. \end{aligned}$$

§ 8.5.2 regression method

- In real application, replacing $\boldsymbol{\mu}$, $\boldsymbol{A}$ and $\boldsymbol{D}$ by $\bar{\boldsymbol{x}}$, $\hat{\boldsymbol{A}}$ and $\hat{\boldsymbol{D}}$ and substitute data of sample $\boldsymbol{x}_j$ in $\boldsymbol{x}$, we obtain the relevant factor score:

$$\tilde{\boldsymbol{f}}_j = \hat{\boldsymbol{A}}' \boldsymbol{S}^{-1} (\boldsymbol{x}_j - \bar{\boldsymbol{x}}) \quad (8.5.11)$$

where $\boldsymbol{S} \approx \hat{\boldsymbol{A}}\hat{\boldsymbol{A}}' + \hat{\boldsymbol{D}}$ in (8.3.2) by principal component analysis method.

$$\tilde{\boldsymbol{f}} = E(\boldsymbol{f} | \boldsymbol{x}) = \boldsymbol{A}' \boldsymbol{\Sigma}^{-1} (\boldsymbol{x} - \boldsymbol{\mu}) \quad (8.5.8)$$

Similarly, from $\tilde{f} = (I + A'D^{-1}A)^{-1} A'D^{-1}(\mathbf{x} - \boldsymbol{\mu})$ in (8.5.10), we obtain the following conditional expectation of factor score $\tilde{f}$:

$$\begin{aligned} E(\tilde{f} \mid f) &= (I + A'D^{-1}A)^{-1} A'D^{-1}E(Af + \varepsilon \mid f) \\ &= (I + A'D^{-1}A)^{-1} A'D^{-1}Af \\ &= (I + A'D^{-1}A)^{-1} (A'D^{-1}A + I - I)f \\ &= f - (I + A'D^{-1}A)^{-1} f \end{aligned} \tag{8.5.13}$$

Average forecast error of factor score $\tilde{f}$ by regression method:

It follows from (8.5.10) that

$$\begin{aligned}\tilde{f} - f &= (I + A'D^{-1}A)^{-1} A'D^{-1} (Af + \varepsilon) - f \\ &= (I + A'D^{-1}A)^{-1} (A'D^{-1}A - (I + A'D^{-1}A))f \\ &\quad + (I + A'D^{-1}A)^{-1} A'D^{-1}\varepsilon \\ &= (I + A'D^{-1}A)^{-1} A'D^{-1}\varepsilon - (I + A'D^{-1}A)^{-1} f.\end{aligned}$$

Then, we have

$$\begin{aligned}&E[(\tilde{f} - f)(\tilde{f} - f)'] \\ &= (I + A'D^{-1}A)^{-1} A'D^{-1}E(\varepsilon\varepsilon')D^{-1}A(I + A'D^{-1}A)^{-1} \\ &\quad + (I + A'D^{-1}A)^{-1}E(ff')(I + A'D^{-1}A)^{-1} \\ &= (I + A'D^{-1}A)^{-1}(I + A'D^{-1}A)(I + A'D^{-1}A)^{-1} \\ &= (I + A'D^{-1}A)^{-1}\end{aligned}\tag{8.5.15}$$

Average forecast error of estimation of $\tilde{f}$

$$E[(\tilde{f} - f)(\tilde{f} - f)'] = (I + A' D^{-1} A)^{-1} \quad (8.5.15)$$

Also, the average forecast error of WLS estimation $\hat{f}$ is

$$E[(\hat{f} - f)(\hat{f} - f)'] = (A' D^{-1} A)^{-1} \quad (8.5.14)$$

It is noticed that (exercise 8.2).

$$(A' D^{-1} A)^{-1} - (I + A' D^{-1} A)^{-1} > 0$$

Therefore, factor scores obtained by regression method are more precise than that by WLS method.

Exercise 8.2. Prove that

$$(A'D^{-1}A)^{-1} - (I + A'D^{-1}A)^{-1} > 0$$

Exercise 8.2. Prove that

$$(A^T D^{-1} A)^{-1} - (I + A^T D^{-1} A)^{-1} > 0$$

Proof: $(A^T D^{-1} A)^{-1} - (I + A^T D^{-1} A)^{-1}$

$$= (I + A^T D^{-1} A)^{-1} [(I + A^T D^{-1} A) - A^T D^{-1} A] (A^T D^{-1} A)^{-1}$$

$$= (I + A^T D^{-1} A)^{-1} (A^T D^{-1} A)^{-1}$$

$$= (A^T D^{-1} A + A^T D^{-1} A A^T D^{-1} A)^{-1}.$$

Then, we can obtain

$$(A^T D^{-1} A)^{-1} - (I + A^T D^{-1} A)^{-1} > 0$$

from $A^T D^{-1} A + A^T D^{-1} A A^T D^{-1} A > 0$.

► **Example 8.5.1** In example 8.4.2, factor scores by regression method are calculated as:

$$\tilde{\mathbf{f}}_j = \hat{\mathbf{A}}' \mathbf{S}^{-1} (\mathbf{x}_j - \bar{\mathbf{x}}) \quad (8.5.15)$$

$$\hat{\mathbf{f}}_{3 \times 1}^* = \hat{\mathbf{A}}_{10 \times 3}^{*T} \hat{\mathbf{R}}_{10 \times 10}^{-1} \mathbf{x}_{10 \times 1}^* \quad \text{see Table 8.4.2 for } \hat{\mathbf{R}}$$

where $\hat{\mathbf{A}}^*$ is defined in Table 8.4.4, $\mathbf{x}^* = (x_1^*, x_2^*, \dots, x_p^*)'$, x_i^* is the standardized value of x_i , $i=1, 2, \dots, p$. After calculation, we obtain

$$\begin{aligned} \hat{f}_1^* = & 0.217x_1^* + 0.216x_2^* + 0.145x_3^* + 0.138x_4^* - 0.054x_5^* \\ & - 0.032x_6^* - 0.066x_7^* - 0.066x_8^* + 0.254x_9^* + 0.246x_{10}^* \end{aligned}$$

$$\begin{aligned} \hat{f}_2^* = & -0.109x_1^* - 0.043x_2^* + 0.116x_3^* + 0.144x_4^* + 0.235x_5^* \\ & - 0.165x_6^* + 0.381x_7^* + 0.371x_8^* - 0.086x_9^* - 0.016x_{10}^* \end{aligned}$$

$$\begin{aligned} \hat{f}_3^* = & 0.100x_1^* + 0.098x_2^* + 0.004x_3^* - 0.037x_4^* + 0.216x_5^* \\ & + 0.876x_6^* - 0.229x_7^* - 0.157x_8^* - 0.008x_9^* - 0.255x_{10}^* \end{aligned}$$



Substitute the standardized value of ten indexes in their financial statement into the above equation to obtain 3 factor scores of each stock. According to the numerical size of factor scores $\hat{f}_1^*, \hat{f}_2^*, \hat{f}_3^*$, order the values from high to low in Table 8.5.1, 8.5.2 and 8.5.3.

Table 8.5.1 Ranked by scale factor score $\hat{f}_1^*$ (规模因子)

number	stock	$\hat{f}_1^*$	$\hat{f}_2^*$	$\hat{f}_3^*$	number	stock	$\hat{f}_1^*$	$\hat{f}_2^*$	$\hat{f}_3^*$
1	上海石化	8.580	-2.704	-2.168	⋮	⋮	⋮	⋮	⋮
2	东方航空	7.446	-2.089	-1.861	595	康美药业	-0.701	0.231	1.624
3	兖州煤碳	6.924	1.513	-0.044	596	潜江制药	-0.706	-0.430	2.085
4	马钢股份	6.175	-1.251	-2.804	597	浏阳花炮	-0.709	0.146	0.655
5	宁沪高速	5.341	0.835	-2.220	598	浪潮软件	-0.713	1.625	-1.313
6	广州控股	4.101	2.596	0.640	599	兆维科技	-0.728	2.511	-1.366
7	青岛海尔	4.022	0.954	3.160	600	PT农商社	-0.751	0.516	0.510
8	四川长虹	3.996	-2.027	1.907	601	三佳模具	-0.776	0.527	0.385
9	仪征化工	3.873	-0.964	-1.598	602	雄震集团	-0.817	1.175	-1.407
10	上海汽车	3.834	1.293	-0.666	603	中软股份	-1.023	2.715	-1.685
⋮	⋮	⋮	⋮	⋮	604	天地科技	-1.023	2.355	-0.946

Table 8.5.2

Ranked by revenue factor score $\hat{f}_2^*$ (盈利因子)

number	stock	$\hat{f}_1^*$	$\hat{f}_2^*$	$\hat{f}_3^*$	number	stock	$\hat{f}_1^*$	$\hat{f}_2^*$	$\hat{f}_3^*$
1	中软股份	-1.023	2.715	-1.685	⋮	⋮	⋮	⋮	⋮
2	广州控股	4.101	2.596	0.640	595	东方电机	-0.246	-3.212	-0.385
3	广汇股份	0.517	2.534	-1.608	596	ST嘉陵	-0.144	-3.570	-0.284
4	兆维科技	-0.728	2.511	-1.366	597	ST海药	-0.089	-3.709	0.225
5	长江通讯	-0.657	2.369	1.899	598	鼎天科技	0.034	-4.230	-0.209
6	天地科技	-1.023	2.355	-0.946	599	大元股份	0.111	-4.559	0.284
7	申能股份	3.248	2.158	-0.498	600	新城B股	-0.080	-4.687	-0.086
8	上港集箱	2.992	2.112	1.624	601	银鸽投资	-0.063	-4.869	-0.086
9	中远航运	-0.588	1.957	-1.449	602	济南百货	0.083	-4.968	0.012
10	创业环保	0.797	1.755	-2.099	603	ST东锅	0.263	-5.979	0.272
⋮	⋮	⋮	⋮	⋮	604	国嘉实业	0.491	-7.730	1.055

Table 8.5.3 Ranked by per share factor score $\hat{f}_3^*$ (每股价值因子)

number	stock	$\hat{f}_1^*$	$\hat{f}_2^*$	$\hat{f}_3^*$	number	stock	$\hat{f}_1^*$	$\hat{f}_2^*$	$\hat{f}_3^*$
1	贵州茅台	0.877	1.366	5.750	⋮	⋮	⋮	⋮	⋮
2	用友软件	-0.581	-0.061	5.165	595	PT宝信	-0.571	1.145	-1.760
3	亿阳信通	-0.523	0.124	4.059	596	东方航空	7.446	-2.089	-1.861
4	华泰股份	-0.224	0.061	3.420	597	ST成量	-0.525	0.042	-1.873
5	太太药业	0.047	0.747	3.234	598	ST自仪	-0.185	-0.012	-1.905
6	赣粤高速	0.206	0.100	3.178	599	创业环保	0.797	1.755	-2.099
7	青岛海尔	4.022	0.954	3.160	600	上海石化	8.580	-2.704	-2.168
8	美克股份	-0.699	0.088	2.752	601	山东基建	2.275	0.797	-2.180
9	宇通客车	-0.264	0.604	2.619	602	ST中纺机	-0.390	0.278	-2.182
10	东方通讯	2.401	-0.750	2.593	603	宁沪高速	5.341	0.835	-2.220
⋮	⋮	⋮	⋮	⋮	604	马钢股份	6.175	-1.251	-2.804



Assignment: Assignment has been published on BB.

